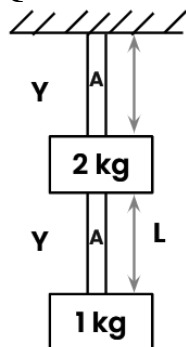


JEE-Main-01-02-2024 (Memory Based)
[EVENING SHIFT]

Physics

Question: Find Ratio of Strains in wire 1 & 2



Options:

- (a) 3:1
- (b) 1:4
- (c) 1:3
- (d) 4:1

Answer: (a)

Question: 2 trains run on N-S track. Train moves with a velocity 20 m/s towards north and train (B) moves with a velocity 30 m/s towards south.

Options:

- (a) $-50\hat{j}$
- (b) $50\hat{j}$
- (c) $-30\hat{j}$
- (d) $30\hat{j}$

Answer: (a)

Question: Find the ratio T_i/T_{ii} of time periods of the two pendulums shown.



Options:

- (a) 1
- (b) $\frac{1}{2}$

- (c) 2
(d) $\frac{1}{4}$

Answer: (a)

Question: A solid sphere is rolling purely with speed v on horizontal surface. It rolls up an incline surface and stops at height h . Then height h is
[g is the acceleration due to gravity]

Options:

- (a) $\frac{3}{10} \frac{v^2}{g}$
(b) $\frac{7}{10} \frac{v^2}{g}$
(c) $\frac{5}{7} \frac{v^2}{g}$
(d) $\frac{7}{5} \frac{v^2}{g}$

Answer: (b)

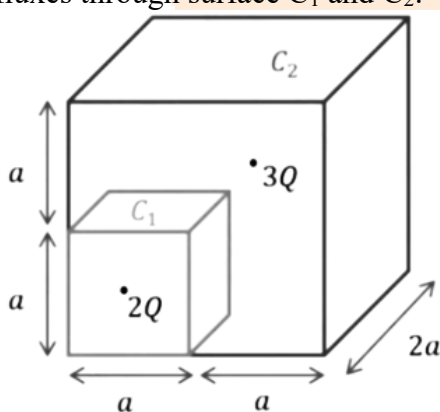
Question: If the power of a light source is P and frequency of photons emitted is f . Find number photons emitted in time t .

Options:

- (a) $\frac{Pt}{2hf}$
(b) $\frac{Pt}{hf}$
(c) $\frac{1}{2} \frac{pf}{ht}$
(d) $\frac{Pf}{ht}$

Answer: (b)

Question: There are two cubical Gaussian surface carrying charges as shown. Find ratio of fluxes through surface C_1 and C_2 :



Options:

- (a) 1 : 1
(b) 2 : 5
(c) 5 : 2
(d) 2 : 3

Answer: (b)

Question: Find the number of significant digits in the value 10.05

Options:

Answer: (4)

Question: $\vec{F}_1 = 6\hat{i} + 3\hat{j} + 1\hat{k}$, $\vec{F}_2 = 2\hat{i} + 1\hat{j} + 3\hat{k}$ Act on 4kg. Find $\vec{a}$

Options:

(a) $\vec{a} = 2\hat{i} + 1\hat{j} + 1\hat{k}$

(b) $\vec{a} = 2\hat{i} - 1\hat{j} + 1\hat{k}$

(c) $\vec{a} = 2\hat{i} - 1\hat{j} - 1\hat{k}$

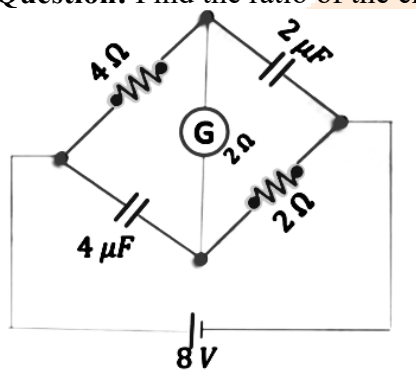
(d) $\vec{a} = 2\hat{i} + 1\hat{j} - 1\hat{k}$

Answer: (a)

Question: A ball of mass 120 g moving with initial velocity 25 m/s is stopped by an external force F in 0.15 sec. Find value of F in newton:

Answer: (20)

Question: Find the ratio of the charge on $4\ \mu\text{F}$ to that on $2\ \mu\text{F}$ in steady state.



Options:

Answer: (3)

Question: A source produces electromagnetic waves of frequency 60 MHz. Find the wavelength of this wave in air.

Options:

Answer: (5)

Question: Efficiency of transformer is 80%, Where input voltage is 10v and input power is 20W in primary. Find current in secondary when voltage is 240 V

Options:

(a) $5.67 \times 10^{-2}\text{ A}$

(b) $4.67 \times 10^{-2}\text{ A}$

(c) $5.67 \times 10^{-2}\text{ A}$

(d) $6.67 \times 10^{-2}\text{ A}$

Answer: (d)

Question: In an isobaric process work done by the gas is 200 J. Adiabatic constant for the gas is 1.4 then find the heat supplied to the gas in the process.

Options:

(a) 500 J

(b) 600 J

(c) 700 J

(d) 800 J

Answer: (c)

Question: 1000 drops of surface energy E_1 , coalesce to form one bigger drop of surface energy E_2 . Find the value of E_2/E_1

Options:

- (a) 1:10
- (b) 1:9
- (c) 1:100
- (d) 100:1

Answer: (a)

Question: A charged particle suspended by thread 20 cm mass is 2 gm and horizontal $E = 2 \times 10^4 \text{ N/C}$. if charge is $1/\sqrt{x} \mu\text{C}$ then find x

Options:

Answer: (3)

Question: Gravitational force of attraction depends on radius of circular motion performed by satellite find dependence of time period on radius

Options:

- (a) $T^2 \propto r^{1/2}$
- (b) $T^2 \propto r^{3/2}$
- (c) $T^2 \propto r^{5/2}$
- (d) $T^2 \propto r^{7/2}$

Answer: (c)

Question: Resistance of Galvanometer is G which is converted into Ammeter. If 0.5% of total current passes through the galvanometer, find resistance of Ammeter.

Options:

- (a) $G/100$
- (b) $G/200$
- (c) $G/199$
- (d) $200G$

Answer: (b)

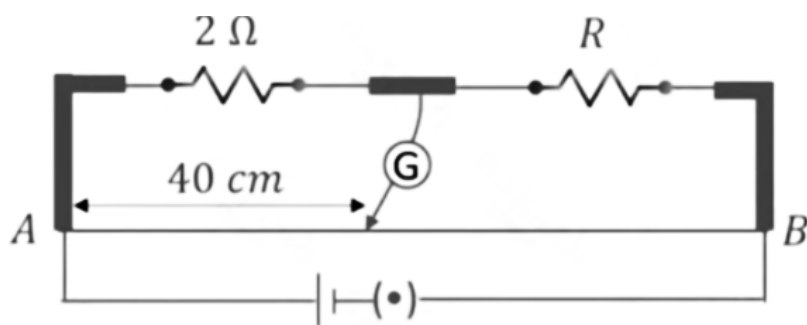
Question: If the rms velocity of hydrogen gas molecules is V_0 , find the rms velocity of oxygen molecules at same temperature:

Options:

- (a) V_0
- (b) $V_0/2$
- (c) $V_0/4$
- (d) $V_0/3$

Answer: (c)

Question: In the meter bridge shown below, the null point is at 40 cm from A. If R is shunted by 2Ω , find the distance of new balance point from A

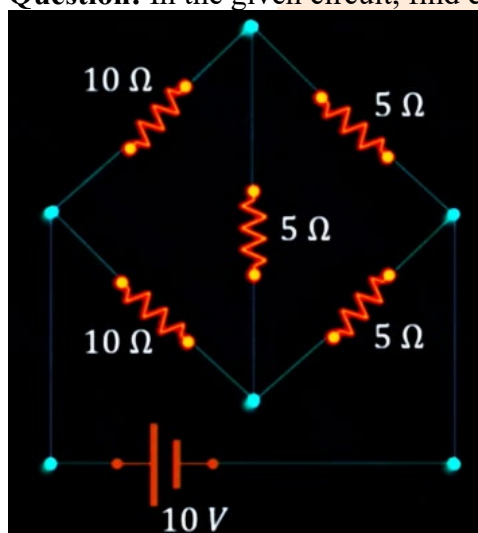


Options:

- (a) 22.7 cm
- (b) 60 cm
- (c) 62.5 cm
- (d) 60.5 cm

Answer: (c)

Question: In the given circuit, find electric current drawn from battery:

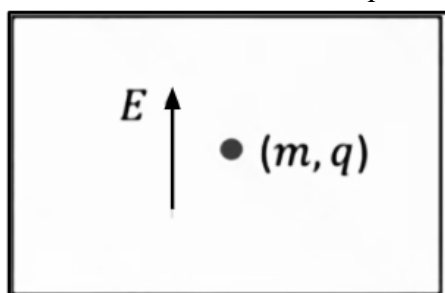


Options:

- (a) $\frac{3}{4}$ A
- (b) $\frac{4}{3}$ A
- (c) $\frac{4}{5}$ A
- (d) $\frac{5}{4}$ A

Answer: (b)

Question: A charged particle (m, q) in vertical plane stays in equilibrium in an electric field as shown. Find the value of q .



Options:

- (a) $\frac{mg}{2E}$
- (b) $\frac{mg}{E}$
- (c) $\frac{2mg}{E}$
- (d) $\frac{mg}{4E}$

Answer: (b)

Question: In Young's double slit experiment, slits separation is 4 cm and separation between slit and screen is 1.5 m. A wave of wavelength 2 cm is incident of slits then find angular width of fringes:

Options:

- (a) 0.75 rad
- (b) 0.65 rad
- (c) 0.80 rad
- (d) 0.5 rad

Answer: (d)

Question: In transition from $n = 2$ to $n = 1$ in hydrogen atom, frequency emitted is f_0 . The frequency for the transition $n = 3$ to $n = 1$ is

Options:

- (a) $\frac{27}{32} f_0$
- (b) $\frac{25}{18} f_0$
- (c) $\frac{32}{27} f_0$
- (d) $\frac{18}{25} f_0$

Answer: (c)

JEE-Main-01-02-2024 (Memory Based)
[EVENING SHIFT]

Chemistry

Question: Number of radial nodes present in 3p are

Options:

- (a) 0
- (b) 1
- (c) 2
- (d) 4

Answer: (b)

Solution: Number of radial nodes = $n - l - 1$

$\therefore$ Number of radial nodes for 3p orbital = $3 - 1 - 1 = 3 - 2 = 1$

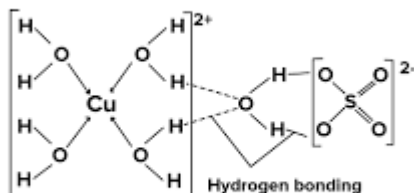
Question: Which of the following compound have colour due to d-d transition ?

Options:

- (a) KMnO_4
- (b) $\text{K}_2\text{Cr}_2\text{O}_7$
- (c) K_2CrO_4
- (d) $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$

Answer: (d)

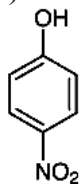
Solution:



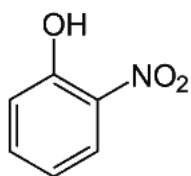
Question: Which of the following compounds has intramolecular hydrogen bonding in it ?

Options:

- (a) NH_3
- (b) H_2O
- (c)

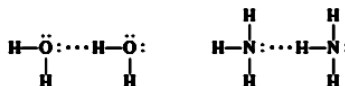
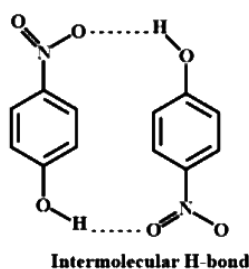
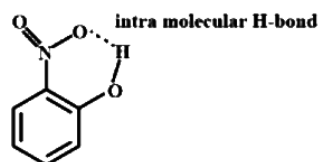


(d)



Answer: (d)

Solution:



Question: Which of the following has highest 3rd ionization energy?

Options:

- (a) Mn
- (b) V
- (c) Cr
- (d) Fe

Answer: (a)

Solution: $Mn = [Ar] 3d^5 4s^2$

Third electrons remove from half filled shell so need more energy to remove this electron.

Question: A 10 mL hydrocarbon ($C_x H_y$) on combustion give 40 mL CO_2 and 50 mL H_2O . Calculate the value of $x+y$

Options:

- (a) 14
- (b) 12
- (c) 11
- (d) 17

Answer: (a)

Solution: The ratio of volumes is $= 10 : 40 : 50 = 1 : 4 : 5$

All the carbon from the hydrocarbon is in the CO_2 . 10 ml of hydrocarbon produces 40ml of CO_2 1 : 4 ratio.

So, 1 mole of hydrocarbon has four moles of carbon.

10 ml of hydrocarbon produces 50 ml water 1 : 5 ratio,

$$x = 4; y = 10$$

$$x + y = 14$$

Question: Solubility of $Ca_3(PO_4)_2$ in 100 mL of pure water is W gm. Find out K_{sp} of $Ca_3(PO_4)_2$ is :

(M : Molecular mass of $\text{Ca}_3(\text{PO}_4)_2$)

Options:

- (a) $108 \times \left(\frac{W}{M}\right)^5$
- (b) $108 \times 10^4 \times \left(\frac{W}{M}\right)^5$
- (c) $108 \times 10^5 \times \left(\frac{W}{M}\right)^5$
- (d) $108 \times 10^6 \times \left(\frac{W}{M}\right)^5$

Answer: (b)

Solution:

The expression for the solubility product of $\text{Ca}_3(\text{PO}_4)_2$ is

$$K_{\text{sp}} = [\text{Ca}^{2+}]^3[\text{PO}_4^{3-}]^2.$$

Substitute values in the above expression.

$$K_{\text{sp}} = \left[3 \times \frac{10 \times W}{M}\right]^3 \left[2 \times \frac{10 \times W}{M}\right]^2 = 108 \left(\frac{10 \times W}{M}\right)^5.$$

Question: Which of the following set of elements can be detected by Lassaigne's test?

Options:

- (a) N and S only
- (b) N, P and S only
- (c) P and halogens only
- (d) N, P, S and halogens

Answer: (d)

Solution: Nitrogen, sulphur, and halogens present in organic compounds are detected by Lassaigne's test. Here, a small piece of Na metal is heated in a fusion tube with the organic compound.

Question: Which of the following compounds in 3d series does not show +3 oxidation state ?

Options:

- (a) V
- (b) Cr
- (c) Mn
- (d) Cu

Answer: (d)

Solution:

Table 8.3: Oxidation States of the first row Transition Metals
(the most common ones are in bold types)

Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
	+2	+2	+2	+2	+2	+2	+2	+1	+2
+3	+3	+3	+3	+3	+3	+3	+3	+2	
	+4	+4	+4	+4	+4	+4	+4		
		+5	+5	+5					
			+6	+6	+6				
				+7					

Question: What is the order of reducing character for AsH_3 , NH_3 , PH_3 (group 15 hydrides)?

Options:

- (a) $\text{NH}_3 > \text{PH}_3 > \text{AsH}_3$
- (b) $\text{PH}_3 > \text{NH}_3 > \text{AsH}_3$
- (c) $\text{AsH}_3 > \text{PH}_3 > \text{NH}_3$
- (d) $\text{NH}_3 > \text{AsH}_3 > \text{PH}_3$

Answer: (c)

Solution:

Reactivity towards hydrogen: All the elements of Group 15 form hydrides of the type EH_3 where E = N, P, As, Sb or Bi. Some of the properties of these hydrides are shown in Table 7.2. The hydrides show regular gradation in their properties. The stability of hydrides decreases from NH_3 to BiH_3 which can be observed from their bond dissociation enthalpy. Consequently, the reducing character of the hydrides increases. Ammonia is only a mild reducing agent while BiH_3 is the strongest reducing agent amongst all the hydrides. Basicity also decreases in the order $\text{NH}_3 > \text{PH}_3 > \text{AsH}_3 > \text{SbH}_3 \geq \text{BiH}_3$.

Question: Assertion: $[\text{Fe}(\text{H}_2\text{O})_5 \text{NO}]\text{SO}_4$ is paramagnetic

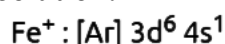
Reason: The Fe in $[\text{Fe}(\text{H}_2\text{O})_5 \text{NO}]\text{SO}_4$ has three unpaired electrons.

Options:

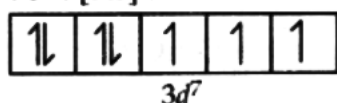
- (a) Both Assertion and Reason are correct and Reason is the correct explanation for Assertion
- (b) Both Assertion and Reason are correct but Reason is not the correct explanation for Assertion
- (c) Assertion is correct but Reason is incorrect
- (d) Both Assertion and Reason are incorrect

Answer: (a)

Solution:



When the weak field ligand H_2O and strong field ligand NO attack, the configuration changes as follows:



$\therefore$ Fe^+ has 3 unpaired electrons.

Question: Consider the following complex $[\text{Co}(\text{NH}_3)_5 \text{CO}_3]\text{ClO}_4$

The coordination number, oxidation number, number of d-electrons and number of unpaired d-electrons on the metal are respectively

Options:

- (a) 6, 3, 6, 0
- (b) 7, 2, 7, 1
- (c) 7, 1, 6, 4
- (d) 6, 2, 7, 3

Answer: (a)

Solution: $[\text{Co}(\text{NH}_3)_5 \text{CO}_3] \text{ClO}_4$. Six monodentate ligands are attached to Co hence C. N of Co = 6

$$\text{O. N} = x + 5 \times (0) + 1 \times (-2) + 1 \times (-1) = 0$$

$\therefore x = +3$; electronic configuration of Co^{3+} [Ar] $3d^6 4s^0$ hence number of d electrons is 6: All d electrons are paired due to strong ligand hence unpaired electrons Zero.



JEE-Main-01-02-2024 (Memory Based)

[EVENING SHIFT]

Mathematics

Question: Let α and β the roots of equation $px^2 + qx - r = 0$, where $P \neq 0$. If p, q, r be the consecutive term of non constant G.P. and $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{3}{4}$, then the value of $(\alpha - \beta)^2$ is:

Answer: $\frac{80}{9}$

Solution:

Question: If the mirror image of the point $P(3, 4, 9)$ in the line $\frac{x-1}{3} = \frac{y+1}{2} = \frac{z-2}{1}$ is (α, β, γ) then find $(\alpha + \beta + \gamma)$ is

Answer: 108.00

Solution:

Question: The number of solutions of the equation $4\sin^2 x - 4\cos^3 x + 9 - 4\cos x = 0$, $x \in [-2\pi, 2\pi]$ is:

Answer: 0.00

Solution:

Question: If the domain of the function $f(x) = \frac{\sqrt{x^2 - 25}}{(\sqrt{4 - x^2})} + \log(x^2 + 2x - 15)$ is

$(-\infty, \alpha) \cup [\beta, \infty)$ then $\alpha^2 + \beta^2$ is equal to

Answer: 50.00

Solution:

Question: Let the system of equation $x + 2y + 3z = 5$, $2x + 3y + z = 9$, $4x + 3y + \lambda z = \mu$ have infinite number of solution. Then $\lambda + 2\mu$ is equal to

Answer: 17.00

Solution:

Question: The value of $\int_0^1 (2x^3 - 3x^2 - x + 1)^{\frac{1}{3}} dx$

Options:

- (a) -1
- (b) 1
- (c) 0
- (d) 2

Answer: (c)

Solution:

Question: The probability that Ajay will not go to office is $\frac{1}{5}$ and probability that Ajay and Vijay will not go to the office is $\frac{2}{7}$, if their visits of office is independent of each other, then find the probability that Ajay will go to the office, but Vijay will not go, is

Options:

- (a) $\frac{12}{28}$
- (b) $\frac{13}{35}$
- (c) $\frac{18}{35}$
- (d) $\frac{24}{35}$

Answer: (c)

Solution:

Question: $\int_0^{\frac{\pi}{3}} \cos^4 x \, dx$ is equal to $a\pi + b\sqrt{3}$, then $a^2 + b$ is equal to:

Options:

- (a) $\frac{1}{2}$
- (b) $\frac{1}{8}$
- (c) $\frac{1}{4}$
- (d) 1

Answer: (b)

Solution:

Question: Let m and n be the coefficient of 7th and 13th term in expansion of

$$\left(\frac{1}{3} x^{\frac{1}{3}} + \frac{1}{2x^5} \right)^{18} \text{ then } \left(\frac{m}{n} \right)^{\frac{1}{3}}$$

Options:

- (a) $\frac{1}{4}$

(b) $\frac{4}{7}$

(c) $\frac{1}{9}$

(d) $\frac{4}{9}$

Answer: (d)

Solution:

Question: The minimum value of $\left|z + \frac{3+4i}{2}\right|; |z| \leq 1$ is

Options:

(a) $\frac{3}{2}$

(b) $\frac{5}{2}$

(c) 3

(d) 5

Answer: (a)

Solution:

Question: Let vertex A(2, 3, 1), B(3, 2, -1), C(-2, 1, 3). If AD is angle bisector of angle A, then projection of $\overrightarrow{AD}$ on $\overrightarrow{AC}$ is equal to

Options:

(a) $\frac{\sqrt{3}}{2}$

(b) $\sqrt{\frac{2}{3}}$

(c) $\sqrt{\frac{3}{2}}$

(d) $\frac{2}{\sqrt{3}}$

Answer: (b)

Solution:

Question: If the system of equations:

$$x + 2y + 3z = 5$$

$$3x + 3y + z = 9$$

$$x + 4y + \lambda z = \mu$$

have infinitely many solutions then the value of $3\lambda + \mu$ equals to

Options:

(a) 17

(b) 21

(c) 43

(d) 34

Answer: (d)

Solution:

Question: $\frac{dx}{dy} = \frac{1+x-y^2}{y}$ and $x(1)=1$, then $5x(2)$ is equal to ____.

Answer: 5.00

Solution:

Question: If $y = \frac{(\sqrt{x}+1)(x^2-\sqrt{x})}{x\sqrt{x}+x+\sqrt{x}} + \frac{1}{15}(3\cos^5 x - 5\cos^3 x)$ then $96y'\left(\frac{\pi}{6}\right)$ equals to ____.

Answer: 105.00

Solution:

Question: Let S_n be the sum of first n terms of an A.P. If $S_{10} = 390$ and the ratio of the tenth and the fifth terms is 15:7, then $S_{15} - S_5$ is equal to

Answer: 790.00

Solution:

Question: There are 20 lines numbered as 1, 2, 3, ..., 20. And the odd numbered lines intersect at a point and all the even numbered lines are parallel. Find the maximum number of point of intersections.

Answer: 101.00

Solution:

Question: Let the focus of the mid point of the chords of the circle $x^2 + (y-1)^2 = 1$ drawn from the origin intersects the line $x + y = 1$ at P and Q. Then the length of PQ is

Answer: $\frac{1}{\sqrt{2}}$

Solution: